MILITARY SPECIFICATION SHEET

INTERCOMMUNICATION SETS, AN/AIC-14 AND AN/AIC-14A

STATION INTERPHONE CONTROL PANEL

C-4951/AIC-14A

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the unit described herein shall consist of this document and the latest issue of specification MIL-I-22352(AS).

1. SCOPE

1.1 Scope.

This specification sheet provides performance and physical characteristics for the station Interphone Control panel C-4951/AIC-14A, hereinafter referred to as C-4951.

1.2 Military Part Number.

The military part number shall be M22352/23-1

2. APPLICABLE DOCUMENTS

Refer to 2.2 of MIL-I-22352(AS) for a listing of applicable documents. The requirements specified herein shall have precedence over all referenced documents. The contract and its attachments shall have precedence over any specification or reference document.

3. REQUIREMENTS

3.1 Item definition. The station Interphone Control panel C-4951 shall provide:

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(1) Interconnection with other interphone stations. One through five interphone stations may be selected at the operators option.

(2) Signal processing consisting of selection and summing of incoming interphone signals.

(3) Visual "in use" indications for the other stations in the IC system.

(4) Capability to monitor, i.e., listen to, the interphone traffic (headset signals) of another station of the ICS.

3.1.1 Item diagram. Station Control Panel C-4951 shall provide the functions represented schematically in accordance with the diagram of Figure 1.

3.1.2 Interface definition. The Station Interphone Control Panel input and output signals will normally display the characteristics presented in Table I.

TABLE I. Interface signal list.

Signal Name	Characteristics
Inputs:	
Interphone No. 1 (INTPH NO. 1)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 1 Vac.
Interphone No. 2 through Interphone No. 5 (INTPH NO. X)	Same as Interphone No. 1
Interphone (INTPH)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 1 Vac.
CALL, Emergency Override (CALL)	Relay logic signal: 1) Open 2) Short to ground
Key Interphone (INTPH KEY)	Relay logic signal: 1) Open 2) Short to ground

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TABLE I. Interface signal list - continued.

Signal Name	Characteristics
Panel Lights (LIGHTS)	0 to ± 28Vdc or 0 to 6V, 400 Hz
Power (PWR)	+27.5 ± 0.5 Vdc
Outputs:	
Mixed Interphone (MIXED INTPH)	Voice range audio (300 to 7000 Hz0 signal ranging in amplitude from 0 to 1 Vac. No dc component normally present.
Common Audio (COMMON AUDIO)	Voice range audio (300 to 7000 Hz0 signal ranging in amplitude from 0 to 1 Vac.
Interphone No. 1 (INTPH NO. 1)	Voice range audio (300 to 7000 Hz0 signal ranging in amplitude from 0 to 1 Vac. No dc component normally present.
Interphone No. 2 through Interphone No. 5 (INTPH NO. X)	Same as Interphone No. 1
Grounds:	
Power (PWR RTN)	Power return. See 3.4.4.2 of MIL-I-22352(AS).
System (SYS RTN)	Signal return. See 3.4.4.3 of MIL-I-22352(AS).

3.1.3 SRAs. The C-4951 shall contain the SRAs listed in Table II.

TABLE II. SRA list.

Title	Application	Detail Spec.
Monitor Amplifier Module	A5101	MIL-I-22352/9(AS)
Lamp Driver Module	A5102	MIL-I-22352/5(AS)

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A brief description of each SRA's function is given below.

3.1.3.1 Monitor Amplifier Module. The Monitor Amplifier Module shall contain:

(1) An audio amplifier used to amplify the interphone audio signals selected by ICS switches placed in the MONITOR position.

(2) An audio amplifier used to amplify the interphone signals selected by ICS switches placed in the SELECTED Position.

(3) A level sensing amplifier which has a 28 Vdc logic output whose state depends upon the amplitude of the audio input.

(4) A monitor amplifier output enable/disable circuit.

The Monitor Amplifier Module shall be used in one place in the C-4951. Performance shall be in accordance with MIL-I-22352/9(AS).

3.1.3.2 Lamp Driver Module. The Lamp Driver Module shall contain six driver circuits, each capable of sinking 100 milliamperes of current sourced from +28 Vdc and passing through a 28 Volt lamp or a +28 Volt relay. The Lamp Driver Module is used in one place within the C-4951 to drive five "in-use" indicating lamps and one Enable Relay within the Monitor Amplifier Module of 3.1.3.1. Performance shall be in accordance with MIL-I-22352/5(AS).

3.1.4 Controls. The Station Interphone Control panel shall provide the controls listed in Table III.

TABLE III. C-4951 controls list.

Title	Function
ICS	The Interphone Control panel shall provide a series of five three position center-off toggle switches as defined herein. The switches shall provide the operator at each station with the capability to select the following functions: MONITOR - In the down position the operator will receive the interphone audio of the station associated with the switch actuated. OFF - In the center position the associated station input is disabled. SELECTED - In the up position the operator can send and receive audio to the selected station when that station has made a complementary selection.

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TABLE III. C-4951 controls list-continued.

Title	Function
ALL-MIXED-SEL'D	The Interphone Control panel shall provide a three position toggle switch with the following functions: SEL'D - In the down position only those signals originating from the station(s) which the operator has selected by setting the corresponding ICS switch(es) shall be presented to the operator for listening. MIXED - In the center position all active traffic on the interphone bus will be amplified and presented to the operator for listening. ALL - In the up position the operator shall hear and transmit to all stations regardless of the settings of the ICS switches at any other station. Stations turned off by setting their respective R/T Interphone Control ON-ICS-OFF switches to the OFF position will not receive the audio.

3.1.5 Indicators. The Station Interphone Control Panel shall provide the indicators listed in Table IV.

TABLE IV. C-4951 Indicator List.

Title	Function
ICS Lamps	The lamp shall light when (1) the associated station is making an attempt to communicate of (2) the associated station is engaged in transmitting traffic. The intensity of the lamp shall vary with the amplitude of the indicated signal.
Panel Lamps	Provide panel illumination.

3.2 Characteristics

3.2.1 Performance characteristics

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3.2.1.1 Input characteristics

3.2.1.1.1 Interphone No. 1 through Interphone No. 5 (INTPH NO. X). The INTPH No. 1 through INTPH No. 5 input characteristics shall be as follows:

3.2.1.1.1.1 Signal Frequency. The INTPH input shall accept drive signals which range from 300 to 7,000 Hz.

3.2.1.1.1.2 Signal Amplitude. The INTPH inputs shall accept drive signals which range in amplitude from 0 to 10 Vac.

3.2.1.1.1.3 Input Impedance. The INTPH inputs shall each present a 10,000 ohm ±10 percent impedance to the incoming signals.

3.2.1.1.1.4 Isolation. The isolation between INTPH signals shall be at least 20 dB for signal ranging from 300 to 7,000 Hz.

3.2.1.1.2 Common Audio (COMMON AUDIO). The COMMON AUDIO line shall act as an output as well as an input. See 3.2.1.2.3 for output characteristics.

The COMMON AUDIO shall be routed via the override relay to the MIXED INTPH output except when the Override relay is energized by a CALL input or alternatively by an INTPH KEY input (ground) when the ALL-MIXED-SEL'D switch is set to the ALL position.

3.2.1.1.2.1 Input Impedance. The COMMON AUDIO input shall each present a 3,300 ohm ±10 percent impedance to the incoming signals.

3.2.1.1.3 Interphone (INTPH).

3.2.1.1.3.1 Signal Frequency. The INTPH input shall accept drive signals which range from 300 to 7000 Hz.

3.2.1.1.3.2 Signal Amplitude. The INTPH input shall accept drive signals which range in amplitude from 0 to 15 Vac.

3.2.1.1.3.3 Input Impedance. The INTPH input impedance shall be 600 ohms ±10 percent.

3.2.1.1.4 CALL, Emergency Override (CALL). The CALL input shall accept a relay drive logic signal having two states: 1) open and 2) short to ground. The CALL input shall be applied to the Override relay which, when energized, connects the INTPH input to the COMMON AUDIO output of Section 3.2.1.2.3.

3.2.1.1.5 Key External (KEY EXT). The KEY EXT input shall accept a relay drive logic signal having two states: 1) open and 2) short to ground. The KEY EXT signal shall be applied to the ALL-MIXED-SEL switch.

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3.2.1.1.6 Power (PWR). The C-4951 shall accept and operate within the constraints of this specification when supplied with +27.5 ±0.5 Vdc electrical power as shown in the diagram of Figure 1. The maximum load presented by the C-4951 shall not exceed 300 milliamperes.

3.2.1.1.7 Panel Lights (LIGHTS). The C-4951 shall be capable of accepting power ranging up to +28 Vdc or 6 Vac at 400 Hz to illuminate the two front panel lamps and five "in use" indicator lamps. The panel lamps used shall be selected accordingly.

TABLE V. Front panel bulbs.

Nomenclature	Power
MS25237A-327 MS25237A-328	+28 Vdc 6 Vac

3.2.1.2 Output Characteristics. All signals which are passed through this unit from input to output by being routed only through connectors, wire, printed wiring, switches, and relays shall not experience insertions loss in excess of 0.5 dB. The C-4951 shall have the following output characteristics:

3.2.1.2.1 Interphone No. 1 through Interphone No. 5 (INTPH NO. X). It shall be possible to select any combination of the INTPH NO. 1 through INTPH NO. 5 outputs to have active signals simultaneously.

With the ICS switch for station NO. 1 in the OFF (Center) or MONITOR (down) position, and the ALL-MIXED-SEL'D switch set to the MIXED position, the INTPH NO. 1 output shall be 28 ±2 dB below a 1.0 Vac at 1,000 Hz input to the INTPH NO. 1 input.

Similarly with the ICS switch for station NO. 1 in the SELECT (up) position and the ALL-MIXED-SEL'D switch in the SEL'D position the output shall be 28 ± 2 dB below a 1.0 Vac at 1,000 Hz input.

With the ICS switch for station NO. 1 in the SELECT (up) position and the ALL-MIXED-SEL'D switch in the MIXED position the output shall be 42 ±2 dB below a 13.9 Vac at 1,000 Hz (+25 dBm) input at the INTPH input.

The switching logic and amplitude of INTPH No. 2 through INTPH No. 5 outputs shall be in accordance with the requirements for INTPH No. 1.

3.2.1.2.2 Mixed Interphone (MIXED INTPH). The MIXED INTPH output shall be the sum of operator selected INTPH No. 1 through INTPH No. 5 inputs. Selection is made by setting the desired ICS switches to the SELECTED (up) position. The MIXED INTPH output amplitude shall be 0 ±4 dB with respect to a 0.03 V, 1,000 Hz input at INTPH NO. 1, INTPH NO. 2, INTPH NO. 3, INTPH NO. 4, or INTPH NO. 5.

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The MIXED INTPH output shall be 44 ±6 dB (87 mV) below a +25 dBm (13.9 volt) input of 1,000 Hz at the COMMON AUDIO input when the ALL-MIXED-SEL'D switch is in the MIXED or ALL position.

3.2.1.2.3 Common Audio (COMMON AUDIO). The COMMON AUDIO shall be connected to the INTPH input of Section 3.2.1.1.1 when the override relay is energized. See 3.2.1.1.2 for COMMON AUDIO input characteristics.

3.2.1.2.4 Indicator No. 1 through Indicator No. 5 (ICS lamp NO. X). The front panel ISC lamp for station No. X shall remain OFF when not enabled by an ICS switch set to MONITOR or SELECT. When enabled by a ground at the INTPH NO. X input, the lamp shall display full brilliance to indicate an attempt of the No. X designated station, e.g., pilot, to communicate. Full brilliance is defined herein as 40 milliampere for +28 Vdc bulbs and 100 milliamperes for 6 Vac bulbs as specified in specification for Lamp Driver Module MIL-I-22352/5(AS).

When enabled and modulated by an audio input from the INTPH NO. X input, the ICS lamp No. X shall indicate by variable intensity that the station is in use and that audio traffic is being transmitted.

3.2.1.2.5 Panel lamps (LIGHTS). The panel lamps shall be edge light type having a red lens in accordance with MIL-L-25467. The C-4951 shall be capable of operating with lamp supply inputs of ranging up to 28 Vdc or 6 Vac at 400 Hz indefinitely without damage.

The unit shall be capable of operating with either ac or dc lighting power source as permitted in MIL-C-6781. When ac power is used for panel lighting self susceptibility to 400 Hz power source shall induce not more than 5 millivolts across any properly'terminated output with the input circuits shorted.

3.2.2 Physical characteristics.

3.2.2.1 Dimensions. The C-4951 Station Interphone Control panel shall consist of a unit having the dimensions of Figure 2. All outside dimensions shall be within ±0.03 inches of the nominal value shown.

3.2.2.2 Weight. The C-4951 shall weigh no more than 1.25 pounds.

3.3 DESIGN AND CONSTRUCTION

The design and construction of the monitor amplifier shall be in accordance with the requirements of 3.3 of MIL-I-22352(AS).

3.3.1 Plastic lighting plates. Plastic lighting plates shall meet the requirements of MIL-P-7788 and Figure 2 of this specification. Markings shown on Figure 2 shall conform with the appropriate application requirements of MS18049. A listing of lighting plates for the C-4951 is contained within Supplement 1 to this specification. If the appropriate configuration drawings are not specified in the contract, this information will be supplied by the procuring activity upon application by the contractor.

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3.3.2 Front panel lighting. The front panel lamps shall be installed using mounting hardware and have red filters in accordance with MIL-L-25467 to provide night vision. The panel lamps shall be mounted in accordance with the outline drawing Figure 2. The lamp drive circuitry shall be capable of accommodating power ranging up to 28 Vdc or 6 V, 400 Hz ac.

4.0 QUALITY ASSURANCE PROVISIONS

See MIL-I-22352(AS).

4.1 Responsibility for tests and inspections. See MIL-I-22352(AS).

4.2 Test Equipment. See 4.3.3.1 of MIL-I-22352(AS).

4.3 Test Methods.

4.3.1 Examination of Product. See 4.3.3.2.1 of MIL-I-22352(AS).

4.3.2 Operational Test. See 4.3.3.2.2 of MIL-I-22352(AS).

4.3.2.1 Input Impedance. Verify that the input impedances are in accordance with 3.2.1.1.1.3, 3.2.1.2.1, 3.2.1.3.3 herein.

4.3.2.2 Gain. Verify that the amplifier gain at 1000 Hz is in accordance with the requirement of 3.2.1.2.1 and 3.2.1.2.2 herein.

4.3.2.3 Logic. Verify the logic requirements of 3.2.1.2.1, 3.2.1.2.2, 3.2.1.2.3, and 3.2.1.2.4 herein.

Preparing Activity:
Navy - AS

Project No. 5831-N042-23

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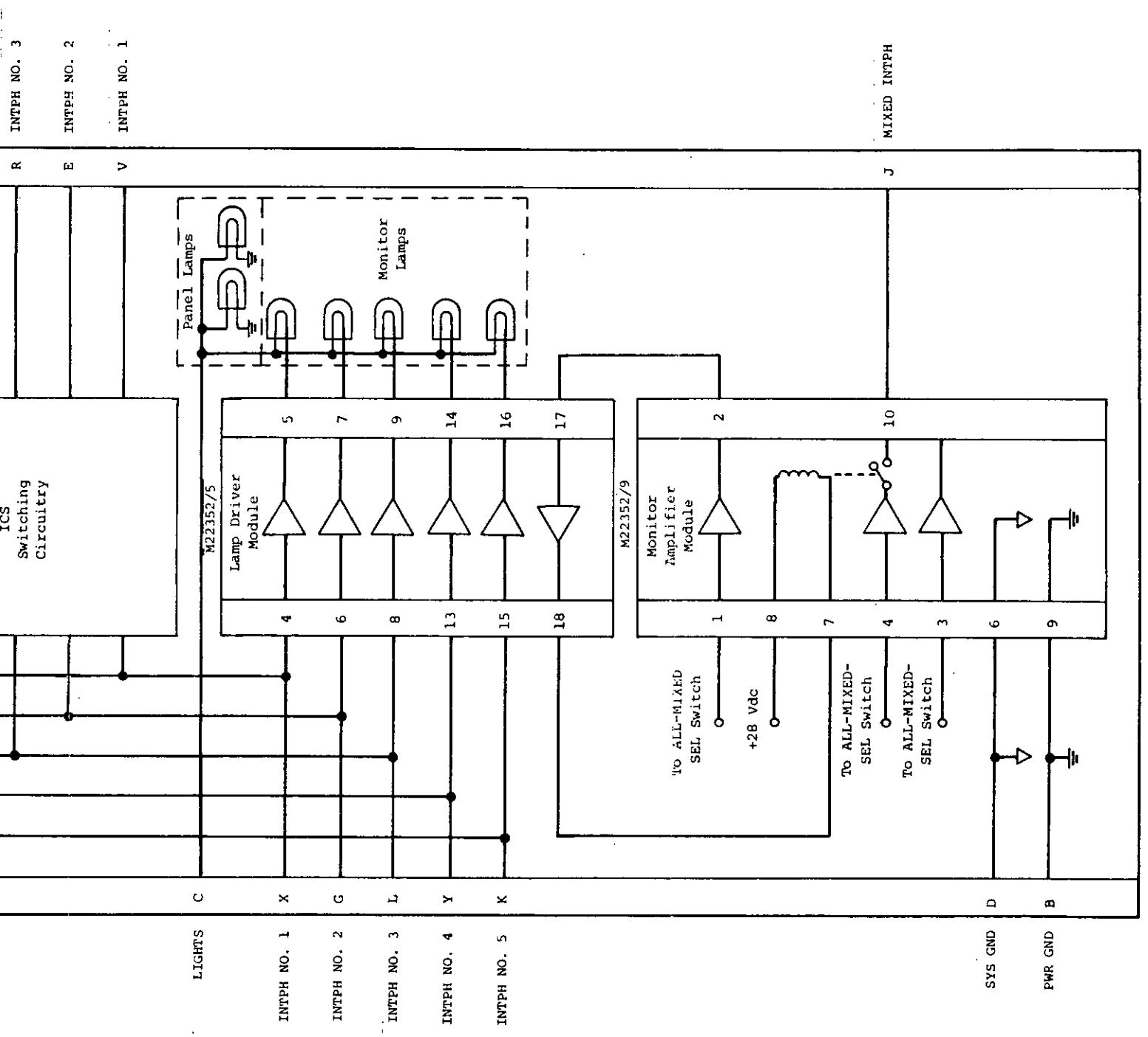
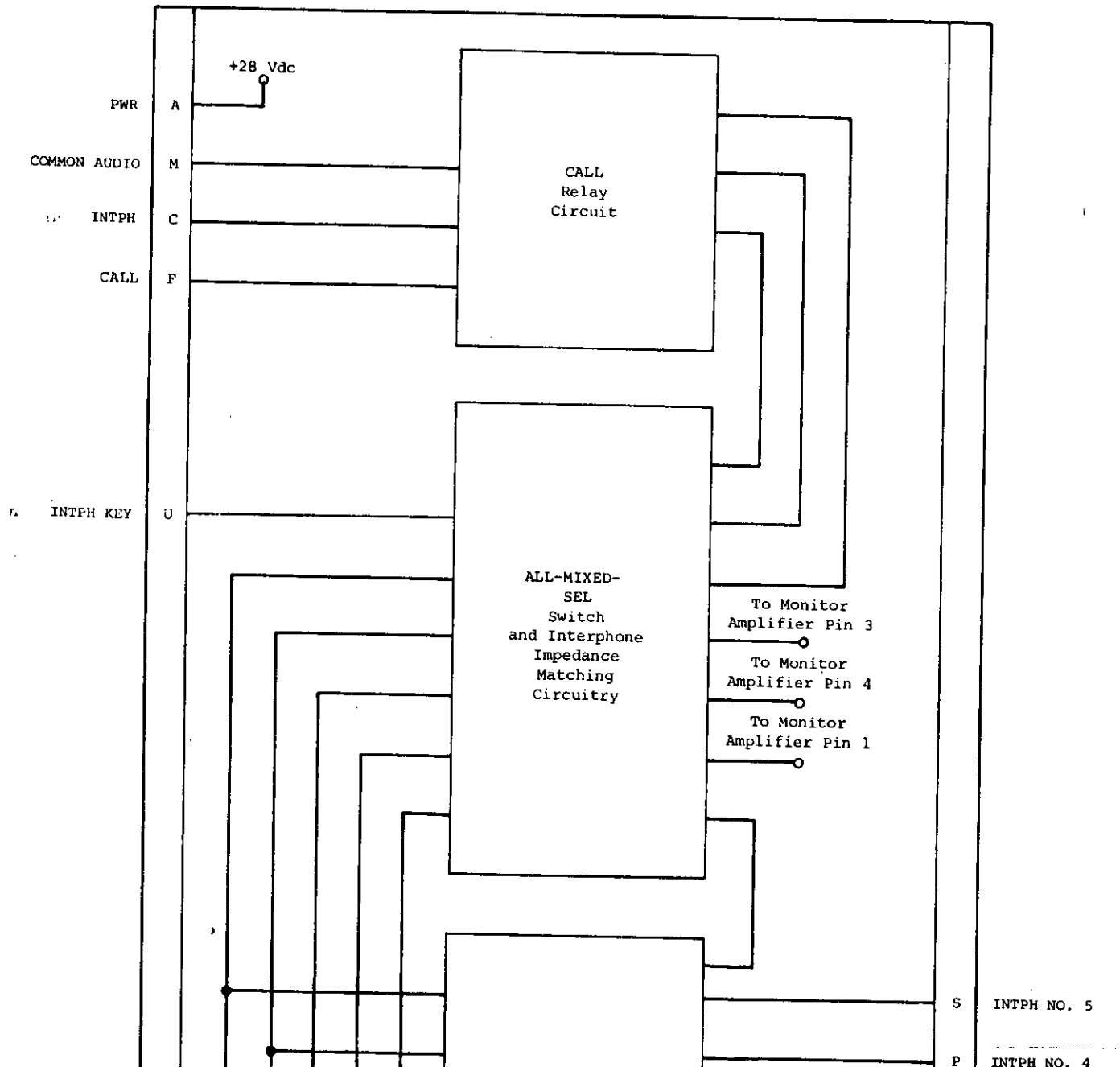


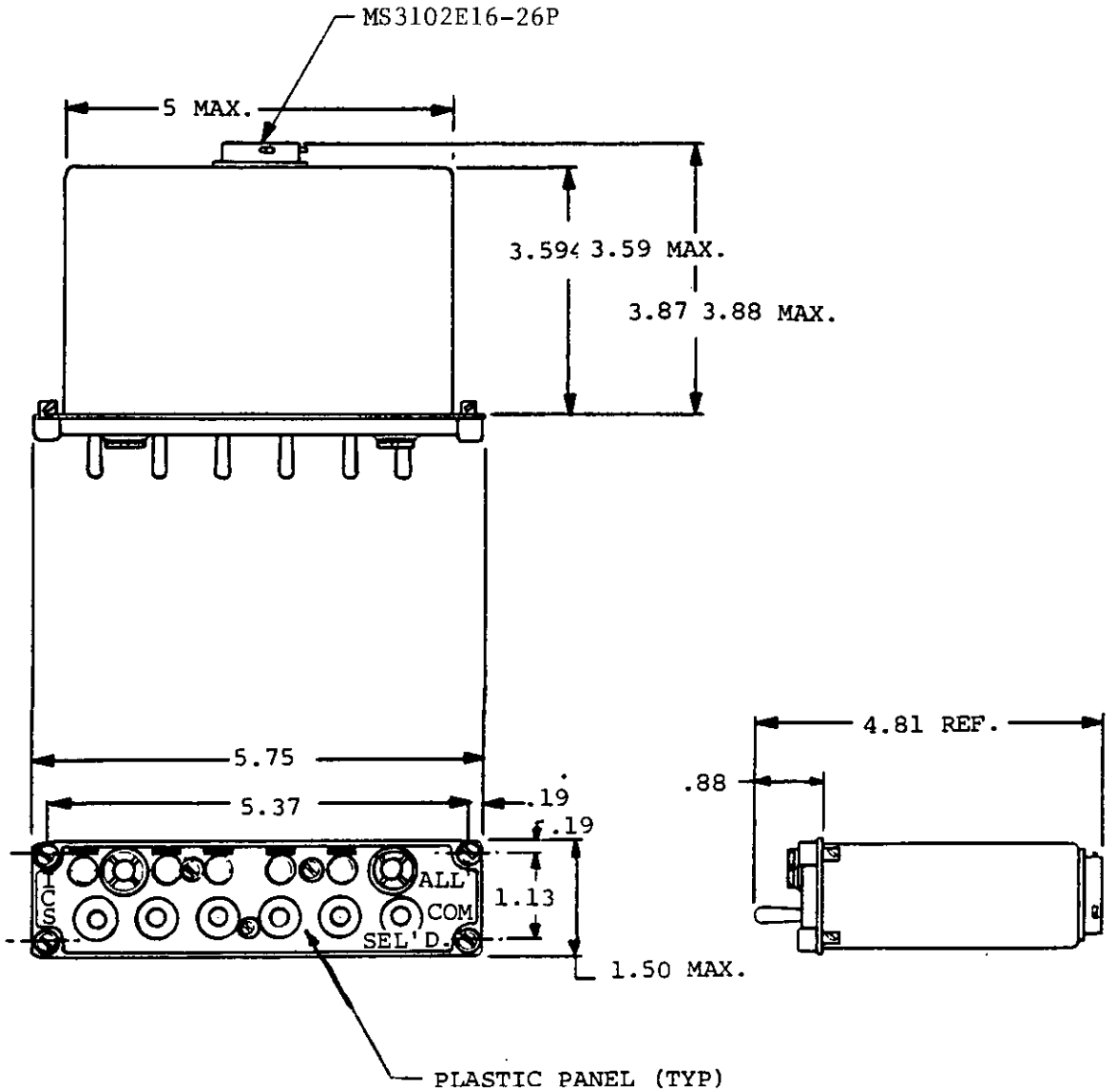
FIGURE 1. C-4951 functional diagram.

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1/ All dimensions are in inches. Tolerances on dimensions are ±0.03 inches unless otherwise specified.

FIGURE 2. C-4951 outline drawing.

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